Maxime Beau, Ph.D.

Systems and Computational Neuroscience

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Google Scholar

Postdoctoral researcher (HHMI, Brody lab, Princeton Neuroscience Institute) studying how brain-wide circuits and catecholaminergic neuromodulation implement attention control during context-dependent decision-making. I combine chronic recordings from up to 8 simultaneous Neuropixels probes in freely moving rats with optogenetics, pharmacology, and dynamical-systems models of multi-area population activity. During my PhD (UCL, Häusser lab), I co-discovered reward-predictive climbing fiber signals in the cerebellum (Kostadinov et al., Nature Neuroscience, 2019) and, as first author in an international collaboration, developed a semi-supervised deep-learning method that identifies neuron cell types across species (Beau et al., Cell, 2025). I also develop open-source Neuropixels software (NeuroPyxels, PixelMap; >200 GitHub stars combined) and co-founded the Princeton Neuroscience Institute Open Science award.

Education

2018-2025 **PhD, Systems and Computational Neuroscience**, *University College London*.
Neural Computation Laboratory. Mentors: Michael Häusser, Beverley Clark, Dimitar Kostadinov

2016-2017 **MSc, Neuroscience**, *Université Pierre et Marie Curie*.
Master Recherche de Sciences, Technologie, Santé - Biologie Intégrative et Physiologie (BIP)

2014-2016 **MD-PhD Programme**, *Université Paris Descartes*.
Programme Médecine-Science

2013-2016 **Medical school Bachelor**, *Université Paris Descartes*.
Diplôme de Formation Générale en Sciences Médicales

Research Positions

2025-present **HHMI Postdoctoral researcher, Neuroscience**.
Brody lab, Princeton Neuroscience Institute, Princeton University, Princeton, NJ, USA. Mentor: Carlos Brody
Projects:
- Brain-wide circuit and neuromodulatory mechanisms of attention control during context-dependent decision-making in freely moving rats.
- Refinement of the 8-probe chronic implant (Bondy et al., 2026) and wiring-aware channelmap design (PixelMap).

Collaborations:
- Stimulant pharmacology in context-dependent decision-making - *Witten lab, Princeton Neuroscience Institute*

2018-2025 **Graduate researcher, Neuroscience**.
Neural Computation Laboratory, University College London, London, UK
Projects:
- Study of long-range information transmission along the cerebellar output pathway in behaving mice.
- Co-discovery of reward-predictive signals in the cerebellum.

Collaborations:
- Co-first author and lead contributor of the Cerebellum Cell-types Classification Collaboration (C4) - *Duke University, Baylor College of Medicine, University College London*
- Testing and deployment of Neuropixels silicon probes - *Neuropixels consortium*
- Comparison of gated linear networks and cerebellar microcircuit architectures - *DeepMind*

Summer 2024 **Trainee, Methods in Computational Neuroscience**.
Marine Biological Laboratory, Woods Hole, MA, USA
Geometrical and dynamical characterisation of RNN activity states during context-dependent perceptual decision-making. Mentored by Dr. Roozbeh Kiani and Dr. Srdjan Ostojic.

2016-2017 **MSc intern, Biophysical Modelling**.
Boris Barbour group, École Normale Supérieure, Paris, France
Computational exploration of the electrogenic properties of the axon initial segment in models of pyramidal and Purkinje cells.

Summer 2016 **Research summer intern, Neural Network Modelling**.
Neural Computation Laboratory, University College London, London, UK
Study of the effect of the topology of artificial networks of cerebellar interneurons on network dynamics.

Summer 2015 **Research summer intern, Neuroendocrinology**.
Franck Oury group, INEM, INSERM U1151, Paris, France

Summer 2014 **Clinical summer intern, Neurosurgery**.
Neurosurgery department, Necker hospital, Paris, France

Selected Publications

2026 **Beau, M.**, Fabre, J., Tabedzki, C., Yanar, J. and Brody, C., *Journal of Open Source Software*.
PixelMap: A browser-based tool for wiring-aware, anatomy- and activity-guided design of Neuropixels channelmaps.

2026 Bondy, A.G., Charlton, J.A., Luo, T.Z., ..., **Beau, M.**, ..., and Brody, C., *bioRxiv*.
Multi-region coordination of evidence accumulation and decision commitment in the brain.

2025 Carolan, J., Land, M.A., Lu, X., **Beau, M.**, ..., and Häusser, M., *Nature Communications*.
All-optical voltage interrogation for probing synaptic plasticity in vivo.

2025 **Beau, M.***, Herzfeld, D.J.*, ..., Häusser, M.† and Medina, J.F.†, *Cell*.
A deep learning strategy to identify cell types across species from high-density extracellular recordings.

2022 Senol, A.D., Pinto, G., **Beau, M.**, ... and Davenne, M., *Brain Communications*.
Alterations of the axon initial segment in multiple sclerosis grey matter.

2021 Sezener, E.*, Grabska-Barwinska, A.*, Kostadinov, D.*, **Beau, M.**, ... and Latham, P., *bioRxiv*.
A rapid and efficient learning rule for biological neural circuits. *UCL, DeepMind*.

2021 Steinmetz, N. A.*, Aydin, C.*, Lebedeva, A.*, Okun, M.*, Pachitariu, M.*, Bauza, M., **Beau, M.** ... and Harris, T., *Science*.
Neuropixels 2.0: A miniaturized high-density probe for stable, long-term brain recordings.

2020 Tsutsumi, S., ..., **Beau, M.** and Häusser, M., *Cell Reports*.
Purkinje cell activity determines the timing of sensory-evoked motor initiation.

2019 Kostadinov, D., **Beau, M.**, Blanco-Pozo, M. and Häusser, M., *Nature Neuroscience*.
Predictive and reactive reward signals conveyed by climbing fiber inputs to cerebellar Purkinje cells.

Awards and Honors

2024 **Jon Driver Prize**, *Winner*, University College London.
Competitive annual prize awarded to outstanding early career neuroscientists from UCL.

2024 **William Morton Wheeler Family Founder's Scholarship**, *Awardee*, Marine Biological Laboratory.
Awarded to attend Woods Hole's summer school: Methods in Computational Neuroscience. $3,000.

2023 **UCL Open Science Award**, *Honourable mention*, University College London.
For NeuroPyxels: open-source Python package for loading, processing and plotting of Neuropixels data.

2015 **Médecine-Science**, *Fellow, Université Paris Descartes*.
MD-PhD programme. €50,400 scholarship, declined. One of 19 laureates.

2014 **Medical School Entry Contest**, *Awardee, Université Paris Descartes*.
Ranked 27th out of 1914 candidates.

Invited Talks

Oct 2026 University of Hong Kong, Croucher workshop: Large-scale Recordings of Neural Activity.

Mar 2026 Cosyne 2026 workshop: from spikes to cell types. Invited speaker.

Jun 2025 Baylor College of Medicine Data Science Community seminar series

Mar 2025 Johns Hopkins Cerebellum Seminars

Jun 2024 Jon Driver Prize lecture, University College London

Jan 2024 French cerebellum day, Institute for Neurosciences of Montpellier, Montpellier, France

Aug 2023 Cerebellum Gordon Research Conference (GRC), Bates College, ME, United States

Jan 2023 French cerebellum day, Institut du Cerveau et de la Moelle Épinière, Paris, France

Jan 2022 Shadmehr lab, Johns Hopkins University, MD, USA

Jan 2020 Systems neuroscience department, Universität Bern, Switzerland

Selected Conference Presentations

2023 - ME, USA **Beau, M.**, Stabb, H., ... and Häusser, M., *Cerebellum Gordon Research Conference*.
What Purkinje cells tell the nuclei: insights from monosynaptic paired recordings in behaving mice.

2022 - CA, USA **Beau, M.***, Herzfeld, D.J.*, ... Medina, J.F.† and Häusser, M.†, *Society for Neuroscience*.
The C4 initiative: Cross-species cell type classification of high-density recordings in the cerebellar cortex.

2019 - Switz. **Beau, M.**, Kostadinov, D. and Häusser, M., *Cerebellum Gordon Research Conference*.
Functional interactions between cerebellar cortex and nuclei in behaving mice.

2018 - CA, USA **Beau, M.**, Kostadinov, D., Blanco Pozo, M. and Häusser, M., *Society for Neuroscience*.
Probing the functional interactions between distinct elements of the cerebellar cortex and deep nuclei circuitry in awake behaving mice.

Open Source Software

- **NeuroPyxels** ★154.
  Python library for electrophysiologists using Neuropixels probes: core utilities for loading, processing and plotting Neuropixels data.
- **PixelMap** ★65.
  Browser-based app for designing anatomy- and activity-guided Neuropixels channelmaps that respect hardware wiring constraints (JOSS, 2026). Hosted at Princeton: pixelmap.pni.princeton.edu.
- **CacheCache** ★13.
  Python library for decorating functions with flexible, runtime-configurable caching.
- **Mplify** ★10.
  Python library for intuitive, one-liner Matplotlib figure prettification.
- **Contributions**.
  Contributor to Kilosort (spike sorting) and Bombcell (automated quality control of spike-sorted units).

Teaching and Mentoring

Teaching.

2026	Woods Hole Methods in Computational Neuroscience	*Teaching assistant, PhDs and postdocs*
2019-present	Paris spring school of Imaging and Electrophysiology	*Teaching assistant, PhDs and postdocs*
2018-present	UCL Neuropixels Course, every year.	*Lecturer, PhDs and postdocs*
2019, 2022	Lisbon Cajal training course: Interacting with neural circuits	*Teaching assistant, PhDs and postdocs*
2018, 2019	UCL Neuroscience BSc: NEURON modelling practicals	*Teaching assistant, undergraduates*

Mentees.

2026-present	Jeremy Schroeter, Princeton Neuroscience Institute PhD student
2021-2023	Hannah Stabb, UCL neuroscience integrated MSci
2021-2022	Federico d'Agostino, UCL machine learning MSc
2020-2021	Ago Lajko, UCL machine learning MSc
2019-2020	Michael Maibach, UCL neuroscience MSc

Technical Expertise

Systems neuroscience.
- *In vivo* high-density electrophysiology: chronic eight-probe Neuropixels implants in freely-moving rats, acute recordings in behaving mice, Neuropixels 1.0 and 2.0 testing (Neuropixels consortium)
- Optogenetics, pharmacology, rodent behaviour (surgery, training, analysis)
- Electronics and signal synchronization for electrophysiology and behaviour, hardware and software
- Confocal microscopy, 3D histological registration

Computational methods.
- Machine learning: clustering, semi-supervised learning, deep generative models (Beau et al., 2025)
- Neural data analysis: spike sorting, spectral analysis, generalized linear models, dynamical systems
- Recurrent neural networks, dynamical systems
- Biophysical modelling: NEURON and BRIAN simulation environments
- Programming: Python (open-source package development), MATLAB, Git

Service and Activities

Society executive positions

2026	Co-founder of the Princeton Neuroscience Institute Open Science Award
2019-2020	Co-founder of UPSyNe, UCL PhDs in Systems Neuroscience association
2019-2022	Executive board member of EMPA, the European MD-PhD association
2017-2018	General Vice President of AMPS, the French MD-PhD association

Editorial and reviewing

2025-present	Recommender, Peer Community In Neuroscience (PCI Neuro); recommendation, 2026
2026	Reviewer, Cosyne

Shotokan Karate Instructor

2025-present	Northern Karate Schools Princeton (northernkarateprinceton.com), Princeton, NJ	*Sensei*
2023-2025	Kentish Town Karate Club, London, UK	*Sempai*
2015	Black Belt 2nd Dan (FFK)	
2013	Black Belt 1st Dan (FFK)	